

# Omar Khaled Ghoz

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[GitHub](#) | [Linkedin](#) | [Portfolio](#)

## AI Engineer | Backend & Data Systems

### Summary:

AI Engineer and Backend Developer with hands-on freelance experience, specializing in building intelligent systems and scalable backend architecture. Experienced in developing Retrieval-Augmented Generation (RAG) pipelines, semantic search engines, and machine learning solutions using Python. Proven ability to design ETL pipelines, integrate **AI** models into production APIs, and build data-driven applications. Backed by solid enterprise customer service experience, I bring a strong understanding of business workflows. Available for full-time opportunities and passionate about solving real-world problems using **AI**, data engineering, and modern backend technologies.

### Work Experience:

#### Freelance Backend & AI/Data Engineer | Independent Clients & Platforms Oct 2022 – Present

- Architected and deployed automated ETL pipelines and advanced Web Scraping scripts using **Python, Playwright, and Selenium**, extracting and structuring over 100,000+ raw data records monthly while reducing data ingestion latency by **40%**.
- Designed and optimized secure, high-performance RESTful APIs using FastAPI and Django, successfully handling up to **10,000 concurrent requests** during peak load periods for scalable client applications.
- Engineered and deployed **5+ Machine Learning models** (Classification, Regression, NLP) into production environments, contributing to an average **18% increase** in client data-driven decision accuracy.
- Automated manual data entry and business processes by integrating backend systems with third-party services (WhatsApp Cloud API, Google Workspace), saving clients an average of **15+ hours** of manual operational work per week.

#### Customer Service & Technical Support Representative | Vodafone & Orange Mar 2023 – Dec 2025

- Delivered exceptional customer and technical support, developing strong communication, problem-solving skills, and a deep understanding of telecommunications business workflows.

### Education:

#### Thebes Academy – Faculty of Business Information System Bachelor of Information Technology | Oct 2022 – Jun 2026

**Relevant Coursework:** Database Design and Management, Big Data Processing Systems, Data Structures and Algorithms, Network Protocols & Web Security, Computer Vision.

### Selected Projects

#### Faragny | AI-Powered Semantic Search & Recommendation Engine

- Architected a full-stack movie recommendation platform replacing traditional keyword search with advanced **Semantic Search** using Vector Databases (ChromaDB) and NLP.
- Developed a high-performance backend using **FastAPI** to handle complex user queries, reducing search latency and significantly improving recommendation accuracy.
- **Tech Stack:** Python, FastAPI, RAG Architecture, ChromaDB, LLMs, Vector Embeddings, **Full-Stack Development**.

#### Orange Telecommunications | Enterprise RAG Customer Service Chatbot

- Engineered a context-aware, zero-hallucination customer support chatbot utilizing **Retrieval-Augmented Generation (RAG)** to answer inquiries based strictly on company-specific documentation.

- Implemented an intelligent **Intent Classification system (Query Analyzer)** to autonomously route users (e.g., tech support, sales) and summarize issues before human-agent handoff.
- **Tech Stack:** Python, LangChain, LLMs, Prompt Engineering, API Integration.

### Task Management RESTful API | Backend Development

- Architected and developed a robust backend system for a Task Management application using **Django and Django REST Framework (DRF)** to efficiently handle complex CRUD operations.
- Implemented secure user authentication and authorization using **JWT (JSON Web Tokens)**, ensuring strict data privacy and role-based access control for individual users.
- Designed relational database models using Django ORM and optimized API performance by integrating advanced query filtering, searching, and pagination features.
- **Tech Stack:** Python, Django, Django REST Framework (DRF), SQLite/PostgreSQL, JWT, Postman.

### Real Estate Price Prediction Pipeline | Machine Learning

- Built an end-to-end ML pipeline to predict housing prices by training and evaluating multiple regression models (Random Forest, SVR, Linear Regression).
- Conducted rigorous Exploratory Data Analysis (EDA), feature engineering, and hyperparameter tuning using Pandas and Scikit-Learn to optimize model accuracy.
- **Tech Stack:** Python, Scikit-Learn, Pandas, NumPy, Data Visualization.

### Skills:

- **Programming Languages:** Python, SQL, JavaScript, TypeScript, C#, C++.
- **Frameworks:** Django, FastAPI / Flask, Node.js (Express), RESTful APIs, Microservices, Asynchronous Programming.
- **Data & Databases:** ETL, Data Modeling, PostgreSQL, MongoDB, MySQL, Redis, Pandas, NumPy.
- **AI & Machine Learning:** TensorFlow, PyTorch, Scikit-learn, NLP, Model Deployment, Data Visualization, MLOps.
- **Cloud & DevOps:** AWS (S3, EC2, Lambda, Sagemaker) / GCP (Cloud Functions), Docker, Kubernetes, Git/GitHub, CI/CD.
- **Soft Skills:** Technical mentoring (TOT), problem-solving, debugging, code review, effective communication.

### Certifications & Professional Courses:

- **IBM Full Stack Software Developer Professional** – IBM (Coursera), 2025
- **ALX Software Engineering with Specialization in Back-End** – ALX Africa, Jan 2025
- **Machine Learning with Python Certification** – free Code Camp, Nov 2024
- **Data Analysis with Python Certification** – free Code Camp, Nov 2024
- **Foundational C# with Microsoft Certification** – free Code Camp, Sep 2024
- **AI for Beginners Course Completion** – HP Life, Jan 2025
- **AI Starter Kit** – ALX Africa, Mar 2025
- **AI Associate Level** – Telecom Egypt, Aug 2023
- **AI Professional Level** – Telecom Egypt, Mar 2026
- **Microsoft Data Engineer** – Digital Egypt Pioneers Program (DEPI), Dec 2025